

EXPERIMENT NO. 3 / 4

Project Scheduling Using PERT/CPM and Gantt Chart

Project: Hospital Management System (HMS)

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1. Aim

To use project management scheduling techniques (PERT/CPM) to determine activity durations, compute float, establish the Critical Path, and draw the Gantt Chart for the Hospital Management System (HMS).

2. Objectives

- To decompose the HMS project into 11 discrete work breakdown activities (A to K).
- To construct an Activity-on-Arrow (AOA) network and perform Forward Pass and Backward Pass computations.
- To determine Earliest Start (ES), Earliest Finish (EF), Latest Start (LS), Latest Finish (LF), and Total Float (TF) for all activities.
- To identify the Critical Path and draw a comprehensive Gantt Chart representing the 47-day development schedule.

3. Activity Breakdown & CPM Computations

Act	Description	Pred	Dur	ES	EF	LS	LF	Total Float	Critical?
A	Requirement Analysis	-	5	0	5	0	5	0	YES (Critical)
B	SRS IEEE 830 Prep	A	4	5	9	5	9	0	YES (Critical)
C	System Design (DFD/UML)	B	6	9	15	9	15	0	YES (Critical)
D	DB Design & Schema	C	5	15	20	15	20	0	YES (Critical)
E	UI/UX Wireframing	C	8	15	23	21	29	6	NO (Float = 6d)
F	Patient & Appt Module	D	9	20	29	20	29	0	YES (Critical)
G	Doctor Consult &	D	8	20	28	21	29	1	NO (Float =

	EHR								1d)
H	Pharmacy & Lab Module	F	7	29	36	31	38	2	NO (Float = 2d)
I	Billing & Payment Module	F	9	29	38	29	38	0	YES (Critical)
J	System Integration & Test	I	6	38	44	38	44	0	YES (Critical)
K	Deployment & Training	J	3	44	47	44	47	0	YES (Critical)

4. Critical Path Analysis

- Minimum Total Project Duration = 47 Days.
- Critical Path Sequence: A → B → C → D → F → I → J → K
- Managerial Implication: Any slippage in critical activities (A, B, C, D, F, I, J, K) will directly delay the final delivery of the project. Non-critical activities (E, G, H) possess slack float that can absorb minor scheduling delays without affecting the 47-day deadline.

5. Gantt Chart Representation

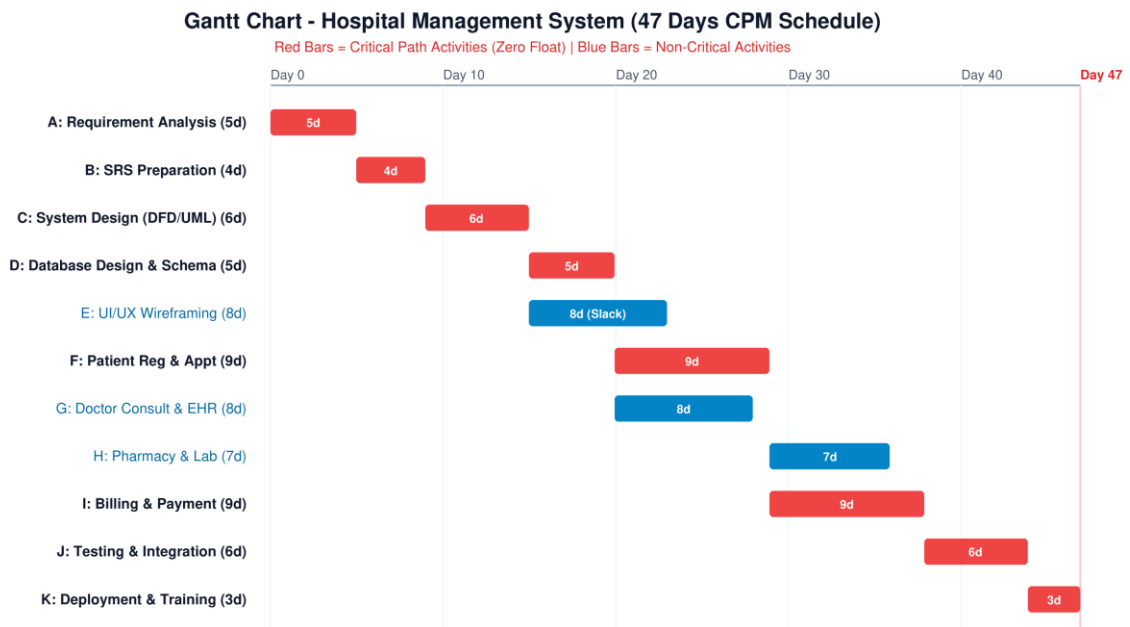


Fig 1: Gantt Chart - Hospital Management System (47 Days CPM Schedule)

6. Conclusion

The PERT/CPM analysis establishes an optimal schedule of 47 working days for the Hospital Management System. The Gantt chart provides a clear visual baseline for monitoring development velocity and mitigating schedule risks.